



Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Getting into Database Structure



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.

Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can explain the concept of database structure.
- I can identify elements that take part in database structure.
- I can give examples of modern database structures.

Ask your partner:

- How do you normally find information organized in a database?
- How do you sort all your data?



Let's Explore!

If you have a data-mess, what you need to use is just a database.



Instructions: Read the text below.

- Jared:** Hey Malcolm! What are you doing?
- Malcolm:** Hey, I'm just here reading some computing stuff to learn. We're having a midterm next Thursday.
- Jared:** What's the midterm about?
- Malcolm:** Database structure. Sorry, we aren't having it on Thursday but on Friday!
- Jared:** I almost never read my notes.
- Malcolm:** Well, you should always read them. A database structure is an organized collection of data, its parts are important for a database structure, some of those parts are: the fields, the record, the table, do you know about it?
- Jared:** I do, I constantly work on that, I'm not forgetting them for the midterm. Fields are information categories that the database stores. In these fields, you add specific categories for your business or company such as: names, addresses, products, etc... It is organized information in a suitable language.
- Malcolm:** It's interesting how a database structure works but you are missing the record and the table. A record is all the information registered for a user in a field; the information is collected in a category and in a field and it could be the record of a person or a thing. A table is when all the records are together; in a few words: a field is an attribute, a record is a collection of attributes and a table is a collection of records.
- Jared:** OK, aren't you missing something? The field type... In a field type, you store any kind of relevant information; numbers, text, dates, memos, yes-no, attachments, etc.
- Malcolm:** The production of a logical structure in the database for example: IDs or registration numbers are used as key fields... A compound key is when a key field is not enough to use. Finally the index, it's a bit that helps database to find records in a faster way.
- Jared:** So, which are the most modern database structures, nowadays?
- Malcolm:** H2, SQLite, ObjectDB and Unqlite that support single file database management.
- Jared:** Well done friend, we are not failing the midterm!

**Let's Practice!**

We are having some presentation next class.

Instructions: Read the text. Write TRUE if the statement is true. Write FALSE if the statement is false.

Now, dear students. I want to talk about our next session. This Thursday we are having two presentations related to the concept of database structure. Milena is presenting a database structure as an organized collection of data and Sebastian is talking about the database structure as a general guide to achieve specific goals. Finally, I am talking about data structure as fundamental creation in computer programming. I suggest you read before attending to the next session. Any questions?

1. Two students are talking about data structure this Monday. ____
2. Milena is presenting database structure as a group of data. ____
3. The instructor is talking about a general guide to achieve specific goals. ____

**Let's Practice!**

Mario usually talks about modern database structures.

Instructions: Select the correct answer to complete the sentences.

1. I _____ manipulate data by using H2 database structure. It has solid security features. (Hint: Adverb that means "commonly".)
 - a. normal
 - b. very
 - c. normally
2. I _____ use SQLite. This is a complete database structure because it contains a single file on disk. (Hint: Not always)
 - a. some
 - b. some of
 - c. sometimes
3. I _____ use the Database Schema Evolution. It is a useful structure, it updates old entity objects in a schema. (Hint: At no time)
 - a. never
 - b. no
 - c. had





Extension Activity

Let's define database structures.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search information about the definition of database structures.
- Write a conversation where you define database structures.
- Include adverbs of frequency and infinitive of purposes.
- Proofread your conversation.
- Share your writing activity with your instructor.

Self-Assessment

Yes

No

- I deepen my knowledge about database structures.
- I express information through a conversation.
- I share my conversation with another person.

☐ ☐
☐ ☐
☐ ☐


Target Words

- **stuff** (noun): information, material or gathered equipment
e.g. Hey, just here reading some computing stuff to learn. We're having a midterm next Thursday.
- **database** (noun): a set of regular and organized data
e.g. It's interesting how a database structure works.
- **information** (noun): a group of collected data
e.g. A record is all the information registered for a user in a field; the information is collected in a category and in a field and it could be the record of a person or a thing.
- **modern** (adjective): something related to the present, use of up to date trends
e.g. So, which are the most modern database structures, nowadays?
- **support** (verb): approval and complement of files and data
e.g. H2, SQLite, ObjectDB and Unqlite that support single file database management.
- **management** (noun): the act of processing and controlling things in general
e.g. H2, SQLite, ObjectDB and Unqlite that support single file database management.



Learning Points

Adverbs of Frequency

Adverbs are words we use to qualify the meaning of other words. They generally indicate **manner** and **frequency**.

Adverbs of Frequency

We can use adverbs of frequency when we want to express the frequency itself, the intensity, the quantity or simply the expression of **how often** we do or perform several activities or **how often** we use something. We can divide **adverbs of frequency** into three categories:

- **Most frequently adverbs**
- **Frequently**
- **Least frequently**

Most frequently adverbs: we use them to express the highest frequency of our activities. For example, we can use some of them such as:

- **Always:** *I **always** use the 'relational model' to find common fields.*
- **Constantly:** *The 'relational model' is **constantly** used to organize field and entities.*
- **Almost always:** *COBOL language **almost always** work to divide all type of data.*

Frequently adverbs: we use them to express frequency in general, not too much, not too little. We can use some of them such as:

- **Frequently:** *Databases **frequently** consist of a single file.*
- **Often:** *Normalization is **often** used to create a set of tables from a set of data files and records.*
- **Sometimes:** ***Sometimes** databases use the customer account number as a common field.*
- **Occasionally:** *Relational databases **occasionally** enforce referential integrity.*

Least frequently adverbs: we use them to express no frequency at all. We can use some of them such as:

- **Never:** *Records **never** reflect changes when they are not updated.*
- **Almost never:** *The customer's record and transactions' record **almost never** get interrelated*
- **Hardly ever:** *Different information **hardly ever** matches in the common field.*

Examples

- *I almost never read my notes.*
- *I do, I constantly work on that.*





Learning Points

Present Continuous

We can use **present Continuous** in several contexts. One of the uses of **present continuous** is to describe plans or events you have for a near future.

You use present continuous for plans in a near future. The negative form follows the same structure:

Subject + Verb To Be (am- are – is) + not + Verb- ing + Complement

Example:

We **aren't missing** the record and the table in the data base structure.

Don't forget to use the verb **Be** according to the subject.

Examples

- *Sorry, we aren't having it on Thursday but on Friday!*
- *I'm not forgetting them for the midterm.*
- *We are not failing the midterm!*